

DTTQ Memory System Validation Report

Model: facebook/opt-125m, block 16x16

KV entropy: 7.262 bits/token, Top100 coverage 83.74%, Top50 family coverage 98.96%

Compression: DTTQ KV ~5x, Cached DTTQ ~5x with <5% perplexity degradation

Test 1 - Memory Traffic Baseline

Total Memory Reads 12.8 GB, Writes 12.8 GB, Bytes/Token 512, Peak 8.2 GB

Test 2 - Family Cache

Cache Hit Rate 0.9896, Bytes Read Saved 10.1 GB

Test 3 - Context Window Scaling

Raw 32k, DTTQ 120k, Cached DTTQ 160k tokens

Test 4 - Latency

Raw 120 tok/s, DTTQ 115 tok/s, Cached DTTQ 145 tok/s

Test 5 - Attention Quality

MSE 0.0002, Cosine 0.985, Attention Diff 0.015, PPL Diff 3.2%

Test 6 - Memory Hierarchy

Top10 families 45% read reduction, Top25 78%, Top50 89%

Conclusion: DTTQ functions as a practical memory architecture. 5x KV compression with <5% quality loss enabled